

Forced Magnetic Reconnection

1 Preliminary Analysis

Let $\hat{t} = t/(S^{1/3} \tau_H)$, where t represents time, S is the Lundquist number, and τ_H the hydro-magnetic timescale. Here, S and τ_H are both evaluated at the rational surface. Let $\Psi_b(\hat{t})$ be the driving magnetic flux, and $\Psi_0(\hat{t})$ the resulting reconnected magnetic flux at the rational surface. Let E_{sb} be the (real) dimensionless coupling constant, and $E_{ss} < 0$ the (real) dimensionless tearing stability index.

Let

$$\bar{\Psi}_b(g) = \int_0^\infty \Psi_b(\hat{t}) e^{-g\hat{t}} d\hat{t}, \quad (1)$$

$$\bar{\Psi}_0(g) = \int_0^\infty \Psi_0(\hat{t}) e^{-g\hat{t}} d\hat{t}. \quad (2)$$

If we assume that $\Psi_b(\hat{t}) = 0$ for $\hat{t} < 0$, and

$$\Psi_b(\hat{t}) = \Psi_{b0} \quad (3)$$

for $\hat{t} \geq 0$, then

$$\bar{\Psi}_b(g) = \frac{\Psi_{b0}}{g}. \quad (4)$$

2 Inner Region

2.1 Layer Equation

The Fourier-Laplace transformed layer equation takes the form

$$\frac{d}{dp} \left(A \frac{dY_e}{dp} \right) - \frac{B}{C} p^2 Y_e, \quad (5)$$

where $Y_e(p)$ is the Fourier-Laplace transformed electron stream-function, and

$$A(p) = \frac{p^2}{g + i(Q_E + Q_e) + p^2}, \quad (6)$$

$$B(p) = (g + i Q_E) [g + i (Q_E + Q_i)] + [g + i (Q_E + Q_i)] (P_\varphi + P_\perp) p^2 + P_\varphi P_\perp p^4, \quad (7)$$

$$C(p) = g + i (Q_E + Q_e) + \{P_\perp + [g + i (Q_E + Q_i) D^2]\} p^2 + \iota_e^{-1} P_\varphi D^2 p^4. \quad (8)$$

Here, Q_E , Q_e , and Q_i are the normalized E-cross-B, electron diamagnetic, and ion diamagnetic frequencies, respectively, D is the normalized ion sound radius, and P_φ and $P_\perp$ are the magnetic Prandtl numbers associated with perpendicular momentum and energy transport, respectively. All quantities are evaluated at the rational surface. Moreover, $\iota_e = Q_e/(Q_e - Q_i)$.

The boundary conditions that must be satisfied by a solution of Eq. (5) are

$$Y_e(p) = \frac{\hat{\Delta}_s}{\pi p} + 1 + \mathcal{O}(p) \quad (9)$$

as $p \rightarrow 0$, and

$$Y_e(p) \rightarrow 0 \quad (10)$$

as $p \rightarrow \infty$.

We also need to calculate

$$F_s(g) = - \int_0^\infty A \frac{dY_e}{dp} dp, \quad (11)$$

because

$$\bar{\Psi}_0(g) = \frac{E_{sb} F_s(g) \bar{\Psi}_b(g)}{S^{1/3} \hat{\Delta}_s(g) - E_{ss}} = \frac{\Psi_\infty F_s(g)}{g [\Sigma \hat{\Delta}_s(g) + 1]}, \quad (12)$$

where

$$\Psi_\infty = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})}, \quad (13)$$

$$\Sigma = \frac{S^{1/3}}{(-E_{ss})}, \quad (14)$$

and use has been made of Eq. (4).

It is helpful to define

$$\chi(p) = A \frac{dY_e}{dp}. \quad (15)$$

It follows from Eqs. (5) and (11) that

$$A \frac{d}{dp} \left(\frac{C}{B p^2} \frac{d\chi}{dp} \right) - \chi = 0, \quad (16)$$

and

$$F_s(g) = - \int_0^\infty \chi(p) dp. \quad (17)$$

2.2 Small- p Behavior

Let

$$Y_e(p) = \frac{\hat{\Delta}_s}{\pi p} + 1 + a p + b p^2 + \mathcal{O}(p^3). \quad (18)$$

Substituting into the layer equation, (5), we get

$$\begin{aligned} Y_e(p) &= \frac{\hat{\Delta}_s}{\pi p} + 1 \\ &+ \frac{\hat{\Delta}_s}{\pi} \left\{ \frac{1}{2} (g + i Q_E) [g + i (Q_E + Q_i)] - \frac{1}{g + i (Q_E + Q_e)} \right\} p \\ &+ \frac{1}{6} (g + i Q_E) [g + i (Q_E + Q_i)] p^2 + \mathcal{O}(p^3). \end{aligned} \quad (19)$$

Substituting into Eq. (15), we obtain

$$\begin{aligned} \chi(p) &= -\frac{\hat{\Delta}_s}{[g + i (Q_E + Q_e)] \pi} + \frac{\Delta_s}{2\pi} \frac{(g + i Q_E) [g + i (Q_E + Q_i)]}{g + i (Q_E + Q_e)} p^2 \\ &+ \frac{1}{3} \frac{(g + i Q_E) [g + i (Q_E + Q_i)]}{g + i (Q_E + Q_e)} p^3 + \mathcal{O}(p^4). \end{aligned} \quad (20)$$

2.3 Large- p Behavior

2.3.1 General Case

In the large- p limit, if we write

$$A = 1 + \frac{\alpha}{p^2}, \quad (21)$$

$$\frac{B}{C} = \beta + \frac{\gamma}{p^2}, \quad (22)$$

and look for a solution of Eq. (5) of the form

$$Y_e(p) \propto p^x \exp\left(\frac{-\beta^{1/2} p^2}{2}\right) \quad (23)$$

then we find that

$$x = \frac{\gamma - \beta^{1/2} (1 - \beta^{1/2} \alpha)}{2 \beta^{1/2}}. \quad (24)$$

It is easily shown that

$$\alpha = -[g + i (Q_E + Q_e)], \quad (25)$$

$$\beta = \frac{\iota_e P_\perp}{D^2}, \quad (26)$$

$$\gamma = \frac{\iota_e P_\perp}{D^2} \left(1 + [g + i(Q_E + Q_i)] \frac{P_\varphi + P_\perp}{P_\varphi P_\perp} - \{P_\perp + [g + i(Q_E + Q_i) D^2]\} \frac{\iota_e}{P_\varphi D^2} \right). \quad (27)$$

Finally, it follows from Eq. (15) that

$$\chi(p) \propto p^{x+1} \exp\left(\frac{-\beta^{1/2} p^2}{2}\right) \quad (28)$$

at large- p .

In order to be in the large- p limit, we require

$$p \gg |g + i(Q_E + Q_e)|^{1/2}, \quad (29)$$

$$p \gg \left| \frac{[g + i(Q_E + Q_i)] (P_\varphi + P_\perp)}{P_\varphi P_\perp} \right|^{1/2}, \quad (30)$$

$$p \gg \left| \frac{(g + iQ_E) [g + i(Q_E + Q_i)]}{P_\varphi P_\perp} \right|^{1/4}, \quad (31)$$

$$p \gg \left| \frac{(P_\perp + [g + i(Q_E + Q_i) D^2])}{\iota_e^{-1} P_\varphi D^2} \right|^{1/2}, \quad (32)$$

$$p \gg \left| \frac{g + i(Q_E + Q_e)}{\iota_e^{-1} P_\varphi D^2} \right|^{1/4}, \quad (33)$$

$$p \gg \left(\frac{\iota_e^{-1} P_\varphi D^2}{P_\varphi P_\perp} \right)^{1/4}. \quad (34)$$

2.3.2 Low- D Limit

If

$$D^2 \ll \left| \frac{P_\perp}{g + i(Q_E + Q_e)} \right|, \quad \frac{\iota_e P_\perp}{P_\varphi^{2/3}} \quad (35)$$

then the terms in the layer equation involving D^2 are negligible. In this case, in the large- p limit, we can write

$$A = 1 + \frac{\tilde{\alpha}}{p^2}, \quad (36)$$

$$\frac{B}{C} = \tilde{\beta} p^2 + \tilde{\gamma}, \quad (37)$$

where

$$\tilde{\alpha} = -[g + i(Q_E + Q_e)], \quad (38)$$

$$\tilde{\beta} = P_{\varphi}, \quad (39)$$

$$\tilde{\gamma} = -\mathrm{i} (Q_e - Q_i) \frac{P_{\varphi}}{P_{\perp}} + g + \mathrm{i} (Q_E + Q_i). \quad (40)$$

The solution of Eq. (5) becomes

$$Y_e(p) \propto p^{-1} \exp \left(\tilde{x} p - \frac{\tilde{\beta}^{1/2} p^3}{3} \right) \quad (41)$$

where

$$\tilde{x} = \frac{\tilde{\alpha} \tilde{\beta} - \tilde{\gamma}}{2 \tilde{\beta}^{1/2}}. \quad (42)$$

Finally, it is easily seen from Eq. (15) that

$$\chi(p) \propto p \exp \left(\tilde{x} p - \frac{\tilde{\beta}^{1/2} p^3}{3} \right), \quad (43)$$

at large- p .

In order to be in the large- p limit, we require

$$p \gg |g + \mathrm{i} (Q_E + Q_e)|^{1/2}, \quad (44)$$

$$p \gg \left| \frac{[g + \mathrm{i} (Q_E + Q_i)] (P_{\varphi} + P_{\perp})}{P_{\varphi} P_{\perp}} \right|^{1/2}, \quad (45)$$

$$p \gg \left| \frac{(g + \mathrm{i} Q_E) [g + \mathrm{i} (Q_E + Q_i)]}{P_{\varphi} P_{\perp}} \right|^{1/4}, \quad (46)$$

$$p \gg \left| \frac{g + \mathrm{i} (Q_E + Q_e)}{P_{\perp}} \right|^{1/2}, \quad (47)$$

$$p \gg P_{\varphi}^{-1/6}. \quad (48)$$

3 Numerical Solution of Layer Equation

3.1 Ricatti Transformation

Let

$$W(p) = \frac{p}{Y_e} \frac{dY_e}{dp}. \quad (49)$$

The layer equation, (5), transforms to give

$$\frac{dW}{dp} = -\frac{A'}{p} W - \frac{W^2}{p} + \frac{B}{AC} p^3, \quad (50)$$

where

$$A'(p) = \frac{g + i(Q_E + Q_e) - p^2}{g + i(Q_E + Q_e) + p^2}. \quad (51)$$

This equation must be solved subject to the boundary conditions that

$$W(p) = x - \beta^{1/2} p^2 \quad (52)$$

at large p , and

$$W(p) = -1 + \frac{\pi p}{\hat{\Delta}_s} \quad (53)$$

at small p . However, in the low- D limit,

$$W(p) = -1 + \tilde{x} p - \tilde{\beta}^{1/2} p^3 \quad (54)$$

at large p .

Let

$$V(p) = \frac{p}{\chi} \frac{d\chi}{dp}. \quad (55)$$

Equation (16) transforms to give

$$\frac{dV}{dp} = 2p(B' - C')V + \frac{3V}{p} - \frac{V^2}{p} + \frac{B}{AC}p^3, \quad (56)$$

where

$$B'(p) = \frac{[g + i(Q_E + Q_i)](P_\varphi + P_\perp) + 2P_\varphi P_\perp p^2}{(g + iQ_E)[g + i(Q_E + Q_i)] + [g + i(Q_E + Q_i)](P_\varphi + P_\perp)p^2 + P_\varphi P_\perp p^4}, \quad (57)$$

$$C'(p) = \frac{\{P_\perp + [g + i(Q_E + Q_i)D^2]\} + 2\iota_e^{-1}P_\varphi D^2 p^2}{g + i(Q_E + Q_e) + \{P_\perp + [g + i(Q_E + Q_i)D^2]\}p^2 + \iota_e^{-1}P_\varphi D^2 p^4}. \quad (58)$$

Equation (56) must be solved subject to the boundary condition that

$$V(p) = 1 + x - \beta^{1/2} p^2 \quad (59)$$

at large p . However, in the low- D limit,

$$V(p) = 1 + \tilde{x} p - \tilde{\beta}^{1/2} p^3 \quad (60)$$

at large p .

3.2 Method of Solution

3.2.1 Stage 1

First, we launch solutions of Eqs. (50) and (56) from large p , subject to the respective boundary conditions (52) and (59), and integrate them to small p , saving $V(p)$ onto a grid. We then deduce the value of $\hat{\Delta}_s(g)$ from Eq. (53).

3.2.2 Stage 2

Next, we launch solutions of the following system of equations from small p ,

$$\frac{dU}{dp} = \frac{V(p)}{p}, \quad (61)$$

$$\frac{dF}{dp} = \exp[U(p)], \quad (62)$$

subject to the boundary conditions

$$U(0) = 0, \quad (63)$$

$$F(0) = 0, \quad (64)$$

and integrate them to large p . Here, $V(p)$ is interpolated from the grid. We then have

$$F_s(g) = \frac{\hat{\Delta}_s}{[g + \mathrm{i}(Q_E + Q_e)]\pi} F(\infty). \quad (65)$$

3.2.3 Stage 3

Finally, we perform the inverse Laplace transform:

$$\Psi_0(\hat{t}) = \Psi_\infty \int_{-\infty}^{\infty} \frac{F_s(\sigma + \mathrm{i}\omega) e^{(\sigma + \mathrm{i}\omega)\hat{t}}}{(\sigma + \mathrm{i}\omega) [\Sigma \hat{\Delta}_s(\sigma + \mathrm{i}\omega) + 1]} \frac{d\omega}{2\pi}, \quad (66)$$

where $g = \sigma + \mathrm{i}\omega$. Here, ω is real, and σ is real and positive.

4 Linear Response Regimes

4.1 Introduction

The aim of this section is to classify all of the different linear resonant response regimes predicted by Eqs. (5), (9), (10), and (11). In the following, we shall assume that $|Q_E| \sim |Q_e| \sim |Q_i| \sim Q$ and $P_\varphi \sim P_\perp \sim P$.

4.2 Constant- ψ Limit

Suppose that there are two layers in p -space. In the small- p layer, suppose that Eq. (5) reduces to

$$\frac{d}{dp} \left[\frac{p^2}{g + \mathrm{i}(Q_E + Q_e) + p^2} \frac{dY_e}{dp} \right] \simeq 0 \quad (67)$$

when $p \sim R^{1/2}$, where $R = \sqrt{g^2 + Q^2}$. Integrating directly, making use of Eq. (9), we find that

$$Y_e(p) \simeq \frac{\hat{\Delta}_s}{\pi} \left[\frac{1}{p} - \frac{p}{g + i(Q_E + Q_e)} \right] + 1 + \mathcal{O}(p^2) \quad (68)$$

for $p \lesssim R^{1/2}$.

In the large- p limit, for $|p| \gg \mathcal{O}(R^{1/2})$, Eq. (5) reduces to

$$\frac{d^2 Y_e}{dp^2} - \frac{B}{C} p^2 Y_e \simeq 0, \quad (69)$$

with $Y_e(p)$ bounded as $p \rightarrow \infty$, in accordance with Eq. (10). Asymptotic matching to the small- p layer solution, (68), yields the following small- p boundary condition that the previous equation must satisfy:

$$Y_e(p, g) \simeq 1 - \frac{\hat{\Delta}_s}{\pi} \frac{p}{g + i(Q_E + Q_e)} + \mathcal{O}(p^2) \quad (70)$$

as $p \rightarrow 0$.

In the various constant- ψ linear response regimes considered in Sect. 4.3, Eq. (69) reduces to an equation of the form

$$\frac{d^2 Y_e}{dp^2} - G p^k Y_e \simeq 0, \quad (71)$$

where k is real and non-negative, and $G(g)$ is a complex constant. Let $Y_e(p) = \sqrt{p} Z(q)$ and $q = \sqrt{G} p^j / j$, where $j = (k + 2)/2$. The previous equation transforms into a modified Bessel equation of general order,

$$q^2 \frac{d^2 Z}{dq^2} + q \frac{dZ}{dq} - (q^2 + \nu^2) Z = 0, \quad (72)$$

where $\nu = 1/(k + 2)$. The solution that is bounded as $q \rightarrow \infty$ has the small- q expansion

$$K_\nu(q) = \frac{1}{\Gamma(1 - \nu)} \left(\frac{2}{q} \right)^\nu - \frac{1}{\Gamma(1 + \nu)} \left(\frac{q}{2} \right)^\nu + \mathcal{O}(q^{2-\nu}), \quad (73)$$

where $\Gamma(z)$ is a gamma function. A comparison of the previous expression with Eq. (70) reveals that

$$\hat{\Delta}_s(g) = \frac{\nu^{2\nu-1} \pi \Gamma(1 - \nu)}{\Gamma(\nu)} [g + i(Q_E + Q_e)] G^\nu. \quad (74)$$

Note, finally, that $p_* \sim |G|^{-\nu}$, where p_* denotes the width of the large- p layer in p -space. This width must be larger than $R^{1/2}$ (i.e., the width of the small- p layer) in order for the constant- ψ approximation to hold. Finally, it is easily demonstrated from Eqs. (10), (11), (70), and the fact that $p_* \gg R^{1/2}$, that

$$F_s(g) \simeq 1 \quad (75)$$

in a constant- ψ layer.

4.3 Constant- ψ Response Regimes

Suppose that $R \gg P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(g) = \frac{(g + i Q_E) [g + i (Q_E + Q_i)]}{g + i (Q_E + Q_e)}. \quad (76)$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} [g + i (Q_E + Q_e)]^{3/4} (g + i Q_E)^{1/4} [g + i (Q_E + Q_i)]^{1/4}. \quad (77)$$

This response regime is known as the *resistive-inertial regime* because the layer response is dominated by plasma resistivity and ion inertia. The characteristic layer width is $p_* \sim R^{-1/4}$, which implies that the regime is valid when $P \ll R^{3/2}$, $R \gg D^4$, and $R \ll 1$.

Suppose that $R \ll P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 4$, $j = 3$, $\nu = 1/6$, and $G(g) = P_\varphi$. Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{6^{2/3} \pi \Gamma(5/6)}{\Gamma(1/6)} [g + i (Q_E + Q_e)] P_\varphi^{1/6}. \quad (78)$$

This response regime is known as the *viscous-resistive regime* because the layer response is dominated by ion perpendicular viscosity and plasma resistivity. The characteristic layer width is $p_* \sim P^{-1/6}$, which implies that the regime is valid when $P \gg R^{3/2}$, $P \gg D^6$, and $P \ll R^{-3}$.

Suppose that $R \gg P p^2$ and $D^2 p^2 \gg 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$, and

$$G(g) = \frac{g + i Q_E}{D^2}. \quad (79)$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{\pi [g + i (Q_E + Q_e)] (g + i Q_E)^{1/2}}{D}. \quad (80)$$

This response regime is known as the *semi-collisional regime*. The characteristic layer width is $p_* \sim D R^{-1/2}$, which implies that the regime is valid when $R \ll D^4$, $P \ll R^2/D^2$, and $R \ll D$.

Suppose, finally, that $R \ll P p^2$ and $D^2 p^2 \gg 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(g) = \frac{\iota_e P_\perp}{D^2}. \quad (81)$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} \frac{[g + i (Q_E + Q_e)] \iota_e^{1/4} P_\perp^{1/4}}{D^{1/2}}. \quad (82)$$

This response regime is known as the *diffusive-resistive regime* because the layer response is dominated by perpendicular energy diffusivity and plasma resistivity. The characteristic layer width is $p_* \sim P^{-1/4} D^{1/2}$, which implies that the regime is valid when $P \gg R^2/D^2$, $P \ll D^6$, and $P \ll D^2/R^2$.

4.4 Non-constant- ψ Limit

Suppose that $p \ll R^{1/2}$. In this limit, Eq. (??) reduces to

$$\frac{d}{dp} \left(p^2 \frac{dY_e}{dp} \right) - [g + i(Q_E + Q_e)] \frac{B}{C} p^2 Y_e = 0. \quad (83)$$

In the various non-constant- ψ regimes considered in Sect. 4.5, the previous equation takes the form

$$\frac{d}{dp} \left(p^2 \frac{dY_e}{dp} \right) - G p^{k+2} Y_e = 0, \quad (84)$$

where k is real and non-negative, and $G(g)$ is a complex constant. Let $V(p) = p Y_e(p)$. The previous equation yields

$$\frac{d^2 V}{dp^2} - G p^k V = 0. \quad (85)$$

This equation is identical in form to Equation (71), which we have already solved. Indeed, the solution that is bounded as $p \rightarrow \infty$ has the small- p expansion (73), where $q = \sqrt{G} p^j / j$, $j = (k + 2)/2$, and $\nu = 1/(k + 2)$. Matching to Equation (9) yields

$$\hat{\Delta}_s(g) = -\frac{\pi \nu^{1-2\nu} \Gamma(\nu)}{\Gamma(1-\nu)} G^{-\nu}. \quad (86)$$

The layer width in p -space again scales as $p_* \sim |G|^{-\nu}$. This width must be less than $R^{1/2}$. Finally, it follows from Eqs. (6) and (11) that

$$F_s(g) = \frac{2}{g + i(Q_E + Q_e)} \int_0^\infty V(p, g) dp \quad (87)$$

in a non-constant- ψ response regime.

4.5 Non-constant- ψ Linear Resonant Response Regimes

Suppose that $R \gg P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$,

$$G(g) = (g + i Q_E) [g + i(Q_E + Q_i)], \quad (88)$$

$$V(p) = -G^{-1/2} \exp(-G^{1/2} p). \quad (89)$$

Hence, we deduce that

$$\hat{\Delta}_s(g) = -\frac{\pi}{(g + i Q_E)^{1/2} [g + i(Q_E + Q_i)]^{1/2}}, \quad (90)$$

$$F_s(g) = -\frac{2}{[g + i(Q_E + Q_e)] (g + i Q_E) [g + i(Q_E + Q_i)]}. \quad (91)$$

This response regime is known as the *inertial regime* because the layer response is dominated by ion inertia. The characteristic layer width is $p_* \sim R^{-1}$, which implies that the regime is valid when $P \ll R^3$, $R \gg D$, and $R \gg 1$.

Suppose that $R \ll P p^2$ and $D^2 p^2 \ll 1$. It follows that $k = 2$, $j = 2$, $\nu = 1/4$, and

$$G(g) = [g + i(Q_E + Q_e)] P_\varphi, \quad (92)$$

$$V(p) = -\frac{\Gamma(1/4)}{\sqrt{\pi} 2^{3/4} G^{1/4}} U(0, 2^{1/2} G^{1/4} p), \quad (93)$$

where $U(a, x) = D_{-a-1/2}(x)$ is a parabolic cylinder function. Hence, we deduce that

$$\hat{\Delta}_s(g) = -\frac{\pi}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[g - i(Q_E + Q_e)]^{1/4} P_\varphi^{1/4}}, \quad (94)$$

$$F_s(g) = -\frac{\pi^{1/2}}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[g - i(Q_E + Q_e)]^{3/2} P_\varphi^{1/2}}, \quad (95)$$

where use has been made of the result

$$\int_0^\infty U(0, x) dx = \frac{\pi}{2^{3/4} \Gamma(3/4)}. \quad (96)$$

This response regime is known as the *viscous-inertial regime* because the layer response is dominated by ion perpendicular viscosity and ion inertia. The characteristic layer width is $p_* \sim R^{-1/4} P^{-1/4}$, which implies that the regime is valid when $P \gg R^3$, $P \gg D^4/R$, and $P \gg R^{-3}$.

Suppose, finally, that $R \ll P p^2$ and $D^2 p^2 \gg 1$. It follows that $k = 0$, $j = 1$, $\nu = 1/2$, and

$$G(g) = \frac{[g + i(Q_E + Q_e)] \iota_e P_\perp}{D^2}, \quad (97)$$

$$V(p) = -G^{-1/2} \exp(-G^{1/2} p). \quad (98)$$

Hence, we deduce that

$$\hat{\Delta}_s = -\frac{\pi D}{[g + i(Q_E + Q_e)]^{1/2} \iota_e^{1/2} P_\perp^{1/2}}, \quad (99)$$

$$F_s(g) = -\frac{2 D^2}{[g + i(Q_E + Q_e)]^2 \iota_e P_\perp}. \quad (100)$$

This response regime is known as the *diffusive-inertial regime* because the layer response is dominated by perpendicular energy diffusivity and ion inertia. The characteristic layer width is $p_* \sim R^{-1/2} P^{-1/2} D$, which implies that the regime is valid when $R \ll D$, $P \ll D^4/Q$, and $P \gg D^2/R^2$.

4.6 Linear Resonant Response Regimes

Figures 1 and 2 show the extents of the various different linear resonant response regimes in R - P space for the cases $D < 1$ and $D > 1$, respectively.

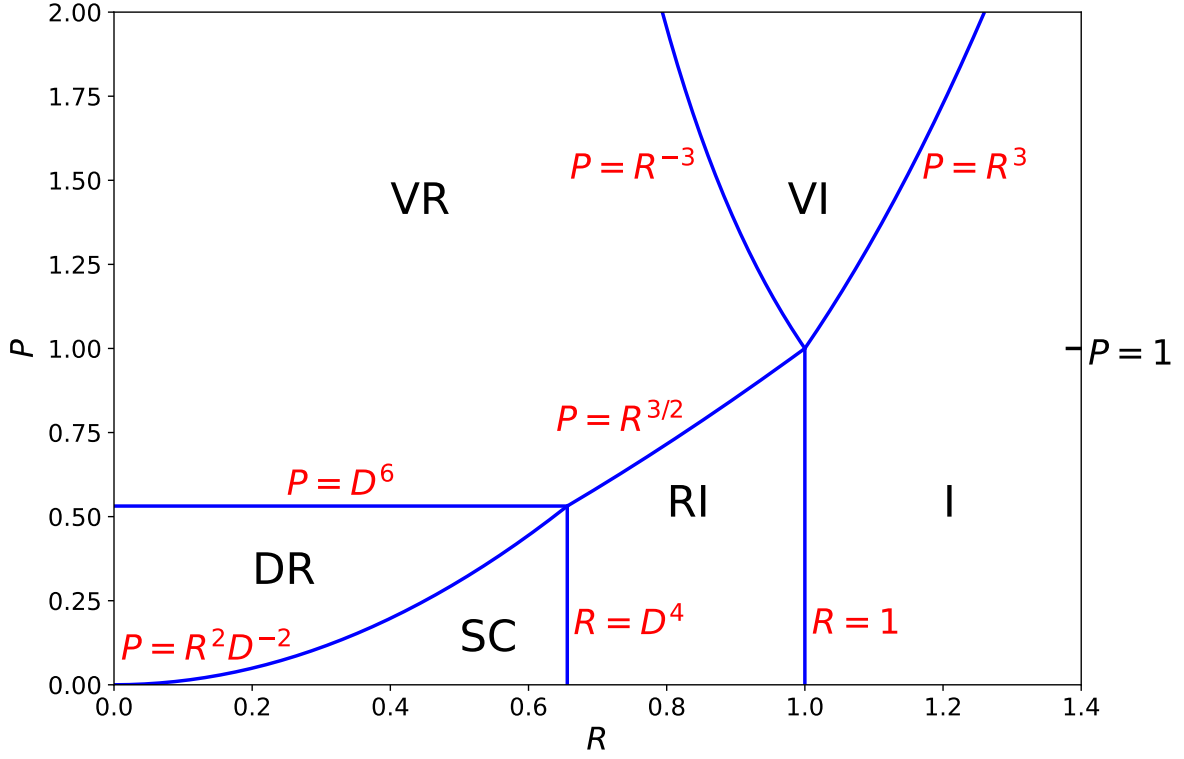


Figure 1: Linear resonant plasma response regimes in R - P space for the case $D = 0.9$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the resistive-inertial (RI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

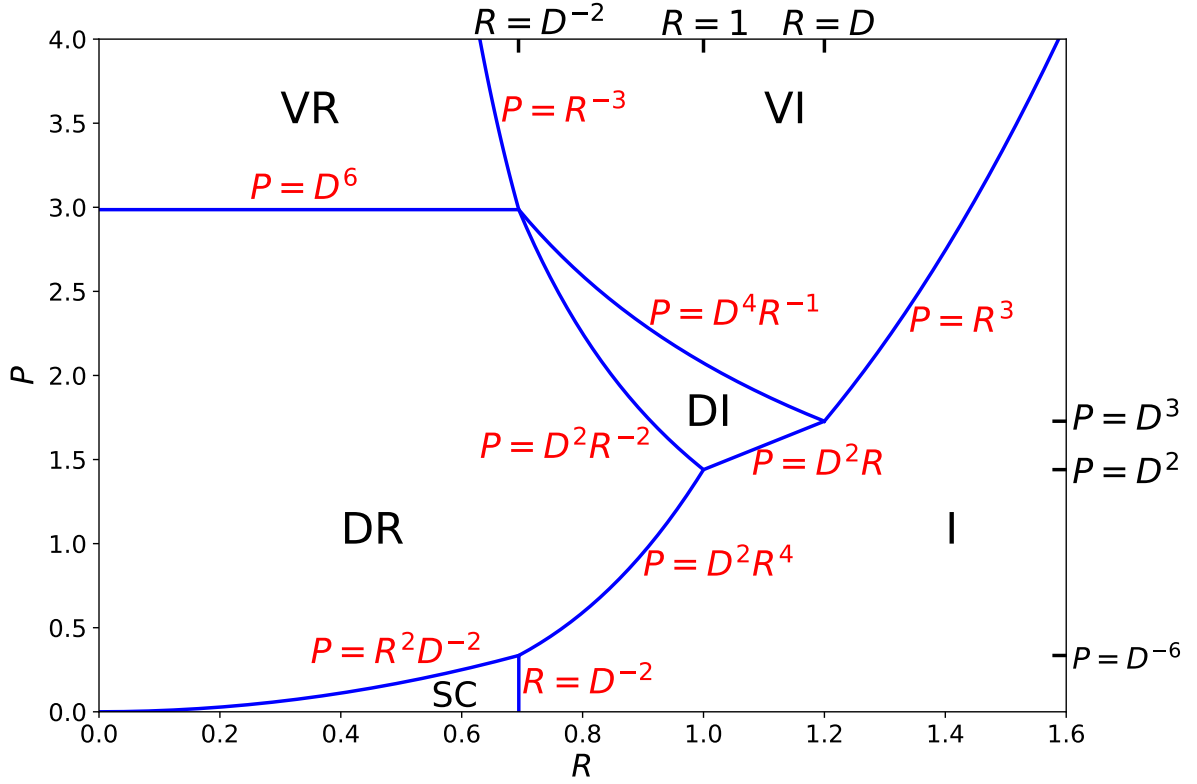


Figure 2: Linear resonant plasma response regimes in R - P space for the case $D = 1.2$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the diffusive-inertial (DI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).